

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: AFRIDOC_MT | **Context:** Amhara Region Agricultural & Microfinance Document Translation — Cooperative and Rural MFI End-Users

Input Ontology definition: Whether the benchmark’s test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark’s task taxonomy (document-level and sentence-level MT between English and five African languages, including Amharic) is structurally relevant — the deployment also needs document-level Amharic MT, and the user agreed that document-level structural demands are broadly similar across genres. However, the test-case category distribution covers only health (WHO) and IT news (Techpoint Africa) genres [Q1, Q2, Q80], with zero coverage of agricultural input subsidy notices, land-use policy updates, credit-scheme terms, or cooperative governance documents — the four genres specified for the deployment. The only legally-binding-text instance in the entire dataset is a translation of the WHO International Health Regulations [DATASET-D17], which does not approach the register of Ethiopian federal proclamations. Web research confirms this is a sector-wide gap: no Amharic MT benchmark covers Ethiopian government proclamation, agricultural subsidy, cooperative governance, or legal-administrative domains [WEB-14, WEB-22, WEB-15]. Given the user’s HIGH priority weight on IO and the total absence of deployment-relevant test categories, the score is low — but not 1, because the abstract task type (document-level Amharic MT) is correctly represented and supports both sentence-level and pseudo-document subtasks at multiple chunk sizes [Q3, Q35, Q49] that broadly match deployment document length variation.

ID	Evidence
Q1	“This paper introduces AFRIDOC-MT, a document-level multi-parallel translation dataset covering English and five African languages: Amharic, Hausa, Swahili, Yorùbá, and Zulu.” (p.1)
Q2	“The dataset comprises 334 health and 271 information technology news documents, all human-translated from English to these languages.” (p.1)
Q3	“We conduct document-level translation benchmark experiments by evaluating the ability of neural machine translation (NMT) models and large language models (LLMs) to translate between English and these languages, at both the sentence and pseudo-document levels, the outputs being realigned to form complete documents for evaluation.” (p.1)
Q35	“Given the AFRIDOC-MT data, we conducted both sentence- and document-level translation, evaluating two types of models: encoder-decoder and decoder-only models. While the majority of these models are open-source, we also evaluated two proprietary models of the same type.” (p.4)
Q49	“To select an appropriate chunk size, we conducted initial tests with k = 1 (sentence-level), 5, 10, and 25, choosing k = 10 for our experiments.” (p.5)
Q80	“We introduce AFRIDOC-MT, a document-level translation dataset in the health and tech domains for five African languages.” (p.9)
D17	አይኤችአር 194 የዓለም ጤና ድርጅት አባላትን ጨምሮ በ196 አገሮች ላይ በሕግ አስፈጻሚ የሆነ የዓለም አቀፍ ሕግ መሣሪያ ነው — Amharic: International Health Regulations — legally-binding international law framework; closest genre to legal-administrative register in the dataset
WEB-14	No public Amharic legal-administrative glossary resource exists (https://github.com/EthioNLP/Ethiopian-Language-Survey)
WEB-15	EthioLLM resources do not include legal-administrative MT data (https://aclanthology.org/2024.lrec-main.561.pdf)
WEB-22	EthioMT parallel corpus is general-domain only (https://arxiv.org/html/2403.19365v1)

Strengths

- Document-level MT task is correctly framed and explicitly motivated by context-dependent phenomena (coreference, lexical disambiguation, lexical cohesion) [Q16] that also matter for cross-clause legal documents
- Multiple chunk-size configurations (k=5, 10, 25, full document, sentence-level) [DATASET-D5, Q49] enable evaluation at granularities relevant to bureau document lengths

- Twelve diverse model configurations evaluated [Q41], including African-centric (Toucan), broad-coverage (NLLB-200, MADLAD-400), and decoder-only LLMs, providing a useful map of MT system options for Amharic

ID	Evidence
Q16	“Both document-level and context-aware MT allow for the possibility of improving translation quality for context-dependent phenomena such as coreference resolution (Müller et al., 2018; Bawden et al., 2018; Voita et al., 2018; Herold and Ney, 2023), lexical disambiguation (Rios Gonzales et al., 2017; Martínez Garcia et al., 2019), and lexical cohesion (Wong and Kit, 2012; Garcia et al., 2014, 2017; Bawden et al., 2018; Voita et al., 2019).” (p.2)
Q41	“We present the result of 12 models in total, including the 1.2B version of Toucan, 1.3B and 3.3B versions of NLLB-200, 3B and 13B versions of MADLAD-400 and Aya-101 respectively. We also have the 8B instruction tuned version of LLama3.1 (LLama3.1-IT), 9B version of Gemma-2 (Gemma2-IT), and LLaMAX3-Alpaca.” (p.4)
Q49	“To select an appropriate chunk size, we conducted initial tests with k = 1 (sentence-level), 5, 10, and 25, choosing k = 10 for our experiments.” (p.5)
D5	ይህ የሚያሳየው የደብሊውፒቪ1 ስርጭት ከአፍጋኒስታን (ምስራቅ ክልል) እና ፓኪስታን (ደቡብ ኬፒ) ዞኖች — Amharic: polio surveillance language, mentions sub-national zones and infection control measures

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required deployment categories are: (1) agricultural input subsidy notices, (2) land-use policy updates, (3) credit-scheme terms / loan agreements, (4) cooperative membership and governance notices — all proclamation-derived formal administrative register documents in Amharic. Hybrid genres (e.g., MFI documents combining credit-scheme and subsidy terminology, since ACSI distributes both [WEB-7]) are also relevant.

IO-2: Yes, the benchmark omits all four required categories. Source corpus is exclusively Techpoint Africa (tech news) and WHO (health news) [Q18, Q22], confirmed empirically across all 61 sampled examples [DATASET-D7, DATASET-D8, DATASET-D9, DATASET-D13]. Web research confirms no other Amharic MT benchmark covers these categories either [WEB-14, WEB-22].

IO-3: The IT/tech news subset (sports betting commentary [DATASET-D28], crypto/Web3 news [DATASET-D9], African tech entrepreneurship interviews [DATASET-D27]) is largely irrelevant to the deployment context and represents construct-irrelevant variance for an agricultural/microfinance use case. Health news has marginal relevance only via its policy-register subset (IHR, TRIPS) [DATASET-D17, DATASET-D18].

IO-4: Documented gaps: (a) total absence of agricultural proclamation register; (b) total absence of cooperative governance documents per Proclamation 985/2016 [WEB-8, WEB-10]; (c) total absence of land-tenure documents per Proclamation 456/2005 and Amhara Revised Proclamation [WEB-12, WEB-13]; (d) total absence of MFI loan-agreement instances. These constitute construct underrepresentation for the deployment's task category.

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Q1	“This paper introduces AFRIDOC-MT, a document-level multi-parallel translation dataset covering English and five African languages: Amharic, Hausa, Swahili, Yorùbá, and Zulu.” (p.1)
Q2	“The dataset comprises 334 health and 271 information technology news documents, all human-translated from English to these languages.” (p.1)
Q18	“We scraped English articles from the websites of Techpoint Africa and the World Health Organization (WHO). The articles cover different topics of different lengths with an average length of 30 and 37 sentences for health and tech respectively.” (p.3)

Q22	"Finally, we selected 334 and 271 English articles/documents from the health and tech domains respectively, which represents 10k sentences each per domain." (p.3)
Q80	"We introduce AFRIDOC-MT, a document-level translation dataset in the health and tech domains for five African languages." (p.9)
D7	ፔይ ደይ ለመሸጥ ይፈልጋል...ናይጂሪያ ፕሬዚዳንት አዲስ CBN ገዥን ሾሙ — Amharic: African technology news digest — IT/fintech domain, no agricultural or legal content
D8	ደቡብ አፍሪካ ጎጂ ይዘቶችን በመስመር ላይ ማቆም ትፈልጋለች...ናላ ከ ዩኬ እና የአውሮፓ ህብረት ለናይጂሪያ ክፍያዎችን ይጀምራል — Amharic: African fintech and content moderation news
D9	ኔስት ኮይን (Nestcoin) የሂሳብ መዛግብቱን ለማጠናከር እና የአዲሱን ምርት በኦንቦርድ እድገት — Amharic: crypto/Web3 fintech startup news
D13	Sahel Capital's SEFAA (Social Enterprise Fund for Agriculture in Africa) Fund — Tech sentence: agriculture-sector investment fund — only agricultural adjacent content in dataset
D17	አይኤችአር 194 የዓለም ጤና ድርጅት አባላትን ጨምሮ በ196 አገሮች ላይ በሕግ አስገዳጅ የሆነ የዓለም አቀፍ ሕግ መሣሪያ ነው — Amharic: International Health Regulations — legally-binding international law framework; closest genre to legal-administrative register in the dataset
D18	ከ ቲአርአይፒኤስ ስምምነት ከፀደቀበት ጊዜ አንሥቶ የዓለም የጤና ስብሰባዎች ብዙ ውሳኔዎች — Amharic: TRIPS agreement, intellectual property law in health sector — international health-law language
D27	ፕሮዳክት ዳይቭ እንዴት ነው የተጀመረው...ወደ ምርት አስተዳደር ጉዞ — Amharic: long-form interview with tech entrepreneur — informal, personal narrative register
D28	In dissecting the social impacts, it becomes vital to delve deep into real-life cases that embody both the positive and negative facets of sports betting in Kenya. — English: sports betting social impacts — not directly relevant to deployment context
WEB-7	https://www.findevgateway.org/sites/default/files/publications/files/mfg-en-paper-subsidizing-microcredit-interest-how-important-is-it-to-the-poor-0-2004.pdf
WEB-8	Cooperative Societies Proclamation 985/2016 defines the operative tier vocabulary (https://leap.unep.org/en/countries/et/national-legislation/cooperative-societies-proclamation-no-9852016)
WEB-10	Ministry of Justice provides authoritative Amharic terminology — federal bureau translators are the preferred authority unrepresented in the benchmark (https://justice.gov.et/en/law/cooperative-societies-proclamation-2/)
WEB-12	Land Administration Proclamation 456/2005 defines land-tenure terminology absent from benchmark (https://land.igad.int/index.php/countries/37-countries/ethiopia/38-ethiopia-profile?start=2)
WEB-13	Amhara Revised Proclamation provides regional land-tenure framework (https://ukgreencitiesandinfrastructure.org/wp-content/uploads/2019/07/ICED-Ethiopia-land-administration-background-paper.pdf)
WEB-14	No public Amharic legal-administrative glossary resource exists (https://github.com/EthioNLP/Ethiopian-Language-Survey)
WEB-22	EthioMT parallel corpus is general-domain only (https://arxiv.org/html/2403.19365v1)

Information Gaps

- Whether deployment intermediaries can compose a custom evaluation set drawing from public Ethiopian proclamation Amharic texts to supplement AFRIDOC-MT
- Whether the deployment also includes oral-mediation use cases (extension worker reading aloud) that would alter the input ontology

Requires Expert Verification

- Confirmation from Amhara Regional Bureau staff that the four listed deployment genres exhaust the operative document categories

Remediation

Gap 1: All four required deployment genres (agricultural input subsidy notices, land-use policy updates, credit-scheme terms, cooperative governance notices) are absent from the benchmark, and no Amharic MT benchmark covers them.

Recommendation: Construct a supplementary evaluation set of 50–200 Amharic deployment-domain documents drawn from publicly available federal/regional proclamations (985/2016, 456/2005, Amhara Revised 2006), EABC subsidy notices, and ACSI loan-agreement templates. Treat AFRIDOC-MT scores as a general-MT baseline only and require deployment-domain test scores for go/no-go decisions.

Gap 2: Hybrid genres (e.g., MFI loan agreements bundled with input-subsidy distribution terms via ACSI's dual government-distribution role [WEB-7]) are not anticipated by either the benchmark or by single-genre supplementary sets.

Recommendation: Include hybrid ACSI-issued documents (MFI + subsidy) in the supplementary deployment-domain test set, sourced via direct consultation with ACSI or Amhara Regional Bureau partners; do not assume single-genre coverage suffices.

ID	Evidence
WEB-7	https://www.findevgateway.org/sites/default/files/publications/files/mfg-en-paper-subsidizing-microcredit-interest-how-important-is-it-to-the-poor-0-2004.pdf